PDA College Of Engineering

Department Of Computer Science and Engineering

SYNOPSIS OF THE PROPOSED MAJORPROJECT (22CS65)

ON

BananaNews

Submitted by:

VTU No.	NAME	SEMESTER	SECTION
3PD23CS050	Hisham Khuram	VI	A
3PD23CS065	Md Ahmed Raza Sofi	VI	A
3PD23CS070	Mohammed Abdul Maaz Hanan	VI	A

Under the Guidance of

Guide Name: Dr. Anita Harsoor

Affiliation : CSE Department, PDA College of Engineering

Academic Year : 2025-26

Department Vision Mission Statement

Vision

To become a premier department in Computer education, research and to prepare highly competent IT professionals to serve industry and society at local and global levels.

Mission

- To impart high quality professional education to become a leader in Computer Science and Engineering.
- To achieve excellence in research for contributing to the development of the society.
- To inculcate professional and ethical behaviour to serve the industry.

ABSTRACT

The rapid growth of digital media has significantly transformed how people access and consume news. Online news platforms have become the primary source of information due to their speed, accessibility, and wide reach. However, this growth has also led to major challenges such as the spread of misinformation, lack of credibility in news sources, and difficulty in delivering relevant content to users. Traditional news websites often focus mainly on publishing content without incorporating intelligent mechanisms for verifying information or improving user engagement through personalization. Therefore, there is a need for a modern news platform that ensures reliable information delivery while also enhancing the overall user experience.

This project proposes the development of **BananaNews**, an intelligent news website that integrates structured news management with concepts inspired by recent research in fake news detection, news recommendation systems, and misinformation analysis. Studies such as the LIAR dataset research highlight the importance of identifying misleading or false information in online news. Similarly, large-scale news recommendation datasets like MIND demonstrate how user behavior and reading patterns can be utilized to recommend personalized news content. Research on misinformation datasets and credibility analysis further emphasizes the need for systems that evaluate the reliability of news sources before presenting them to readers. By studying these approaches, this project aims to design a news platform that prioritizes credibility, organization, and relevance of information.

The proposed system will provide a structured environment where news articles are categorized into different sections such as technology, politics, sports, and entertainment, allowing users to easily navigate through topics of interest. The platform will also incorporate **Artificial Intelligence techniques to assist in verifying the authenticity of news articles, identifying potentially misleading information, and generating short summaries of lengthy news reports for quick understanding**. In addition, concepts from news recommendation research will be used to understand how personalized content delivery can improve user engagement and reading experience. By analyzing user preferences and interaction patterns, the system can highlight trending or relevant news articles for different readers.

The expected outcome of this project is a functional news website that demonstrates how research concepts such as misinformation analysis, news verification, and recommendation techniques can be incorporated into a practical web platform. By combining these approaches with **AI-assisted content verification and summarization**, **BananaNews** aims to contribute toward the development of more reliable, efficient, and engaging online news platforms that address the challenges faced by modern digital journalism.

INTRODUCTION

News plays a critical role in shaping public opinion, spreading awareness, and informing people about events happening around the world. In today's digital era, the internet has become the fastest and most convenient way for people to access news. With smartphones and social media platforms, information can reach millions of readers within seconds. While this speed has improved access to information, it has also created serious challenges related to the accuracy and reliability of the news being shared online.

One of the major concerns in modern digital journalism is the rapid spread of false or misleading information. Unverified news can easily circulate through websites and social media platforms, influencing public perception and sometimes creating panic or confusion. Inaccurate information can affect political opinions, financial decisions, and social harmony. For this reason, ensuring the credibility and authenticity of news content has become an essential requirement for any reliable news platform. Readers increasingly expect news sources to provide accurate information supported by trustworthy sources.

At the same time, the enormous volume of news content published every day makes it difficult for users to keep up with important updates. Many readers prefer short and clear summaries rather than lengthy articles, especially when browsing news on mobile devices. This highlights the need for systems that not only present news in an organized manner but also help users quickly understand key information.

Technological advancements, particularly in Artificial Intelligence, provide new opportunities to improve how news is delivered and verified. AI-based systems can assist in analyzing content, identifying potentially misleading information, and generating concise summaries of longer articles. These technologies can also help organize news content more efficiently and enhance the overall reading experience.

The project **BananaNews** aims to address these challenges by developing a modern digital news platform that focuses on credibility, clarity, and accessibility. By combining structured news organization with AI-assisted analysis and summarization, the platform seeks to provide readers with reliable information while making news consumption faster and more convenient.

EXISTING AND PROPOSED SOLUTIONS

- **ClaimBuster** — an end-to-end claim-spotting system that uses NLP and supervised learning to detect "check-worthy" factual claims in political speech and text.
  ClaimBuster's pipeline focuses on *claim detection* (finding sentences that deserve fact-checking) so human fact-checkers can prioritize work instead of reading every sentence. The academic descriptions show how automatic triage speeds up fact- checking by surfacing the highest-priority claims.

- **Full Fact** — a long-standing fact-checking organisation that builds practical AI tools (Full Fact AI / Live) to monitor public debate, spot likely misinformation in real time, and support human verifiers. Their stack combines claim spotting, source-matching to official datasets, and tooling to track spread and trends — designed to save time for fact-check teams while keeping humans in the loop.

BananaNews is a web platform that *aggregates* reporting from multiple publishers, *automatically assesses* the trustworthiness of incoming items, and *synthesizes* a single coherent article and a short summary for readers. Key modules:

- **Multi-source crawler & deduplication** — collects articles on the same event from many outlets and clusters duplicates/near-duplicates.
- **Claim detection & triage** — spots check-worthy factual claims inside articles (inspired by ClaimBuster).
- **Automated verification / scoring** — ensemble models (fact-check classifiers trained on public datasets) compute a provisional credibility score and surface the most relevant evidence. (Works with human reviewers for borderline/complex cases.) Useful datasets for training include LIAR (fake-news statements) and MIND (news behavior & recommendation research) which inform model design and evaluation.
- **Summarization & synthesis** — produces a concise AI summary ("Short Read") and a consolidated "single big news" article that reconciles multiple sources, notes disagreements, and links to original pieces.

PROBLEM STATEMENT

The rapid growth of online news platforms has made information widely accessible, but it has also increased the spread of misinformation, unverified reports, and misleading headlines.
Many news websites publish content quickly without proper verification, making it difficult for readers to judge the credibility of the information they consume. In addition, users often need to visit multiple websites to understand a single news event, which leads to information overload and inefficient news consumption. These challenges highlight the need for a reliable system that can collect news from multiple sources, verify its authenticity, and present it in a clear and concise format.

METHODOLOGY

1. Data Collection

The first stage of the system involves collecting news articles from multiple trusted online sources and news APIs. These articles will be automatically fetched and stored in a centralized database. Gathering news from different platforms helps provide diverse perspectives and ensures that users receive comprehensive coverage of important events.

2. Data Processing and Categorization

After collecting the news articles, the system will process the content and organize it into different categories such as politics, technology, sports, entertainment, and business. This categorization improves the structure of the website and allows users to easily navigate and access news based on their interests.

3. AI-Based News Verification

Artificial Intelligence techniques will be used to analyze the credibility of news articles. The system will examine the reliability of sources, compare information across multiple reports, and identify potentially misleading or suspicious content. This step helps improve the trustworthiness of the information presented to users.

4. News Aggregation and Content Integration

The platform will gather related articles from multiple news websites and combine their information into a single comprehensive news report. This feature helps users understand a complete story without needing to read several separate articles from different sources.

5. AI-Based News Summarization

An AI summarization module will generate short summaries of lengthy news articles. These summaries will highlight the key points of the news, allowing readers to quickly understand important updates without reading the entire article.

6. Website Development and User Interface

The news platform will be developed using modern web technologies to create a responsive and user-friendly interface. The website will display categorized news, summaries, and detailed articles in a clear and organized format.

7. Testing and Evaluation

The final stage involves testing the system to ensure accuracy, performance, and usability. The website will be evaluated for proper news display, AI functionality, and overall reliability before deployment.

SYSTEM REQUIREMENTS

1. Hardware Requirements

- **Processor:** Intel i3 or higher
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** At least 20 GB free disk space
- **Internet Connection:** Stable broadband connection for accessing news sources,APIs
- **Input Devices:** Keyboard and mouse
- **Output Devices:** Monitor or display screen

2. Software Requirements

- **Operating System:** Windows / macOS / Linux
- **Frontend Technologies:** HTML, CSS, JavaScript
- **Frameworks/Libraries:** Bootstrap / React (optional for UI)
- **Backend Technologies:** Node.js / Python / PHP
- **Database:** MySQL / MongoDB / Supabase
- **AI Tools:** GeminiAPI

SYSTEM SPECIFICATIONS

Reliability: The system should provide accurate and credible news information.

Performance: News articles should load quickly with minimal delay.

Usability: The website interface should be simple and user-friendly.

Scalability: The system should be capable of handling increasing numbers of news articles and users.

Security: Basic security measures will protect stored data and prevent unauthorized access.

EXPECTED OUTCOME OF THE PROJECT

The expected outcome of the **BananaNews** project is the development of a functional and intelligent news website that improves the way users access and understand online news. The platform will provide a structured environment where news from multiple sources is collected, organized, and presented in a clear and user-friendly format. By categorizing news into sections such as technology, politics, sports, and entertainment, users will be able to easily navigate and find information related to their interests.

A key outcome of the system will be the integration of **Artificial Intelligence to assist in improving news reliability and readability**. The platform will analyze news content from different sources and combine related articles into a single comprehensive report. This will help readers understand the complete story of an event without needing to visit multiple websites. The system will also highlight the original sources of information, helping users identify where the news is coming from and improving transparency.

Another important outcome is the implementation of **AI-based summarization**, which will generate short summaries of lengthy news articles. This feature will allow users to quickly grasp the main points of a news story, saving time and making news consumption more efficient, especially for readers who prefer brief updates.

Overall, the BananaNews platform aims to provide a **more reliable, efficient, and accessible digital news experience**. By combining news aggregation, AI-assisted credibility analysis, and automated summarization, the system will demonstrate how modern technologies can be used to reduce misinformation, improve content clarity, and enhance the overall user experience in online journalism. The project will serve as a foundation for further advancements in intelligent news platforms and AI-driven media systems.